Surveilling Surveillance: Estimating the Prevalence of Surveillance Cameras with Street View Data

Hao Sheng, Keniel Yao, Sharad Goel

(2021)

Presented by Hannah Osswald

23.10.2026

ETH zürich

Motivation/Method

Goal: Large scale estimation of number and location of surveillance cameras

- Assemble Street View data
- Train model on manually labeled images
- Estimate city-wide camera count for several cities using the model



1

[1] By Myotus - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=137843478

Data

- Google Street View
- 16 large cities (10 US, 2 Europe, 4 Asia)
- $N = 100'000$ Street View points for every city
- View perpendicular to house facade
- Estimated covered proportion of road network:
  $\text{coverage} = \frac{\#\text{samples} * \text{image_coverage_length}}{2 * \text{road_network_length}}$



2



e.g. New York: $16'362$km roads - ~ 29m/image $\Rightarrow 9\%$ coverage

[2] https://www.google.com/streetview/

Model

- Binary image segmentation
- Training via human labeled data
- Recall $= \frac{\text{TP}}{(TP+FN)}$
- Training data from San Francisco: Recall $= 0.63$

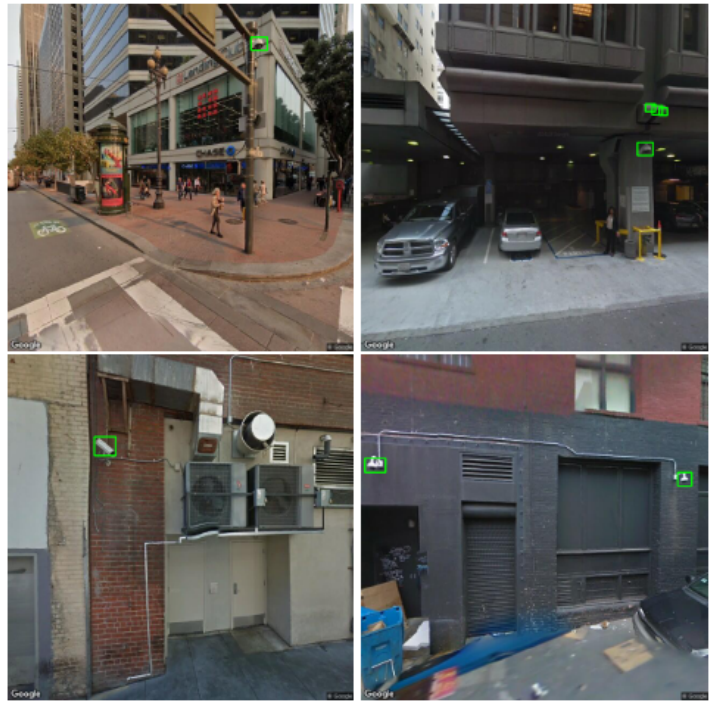


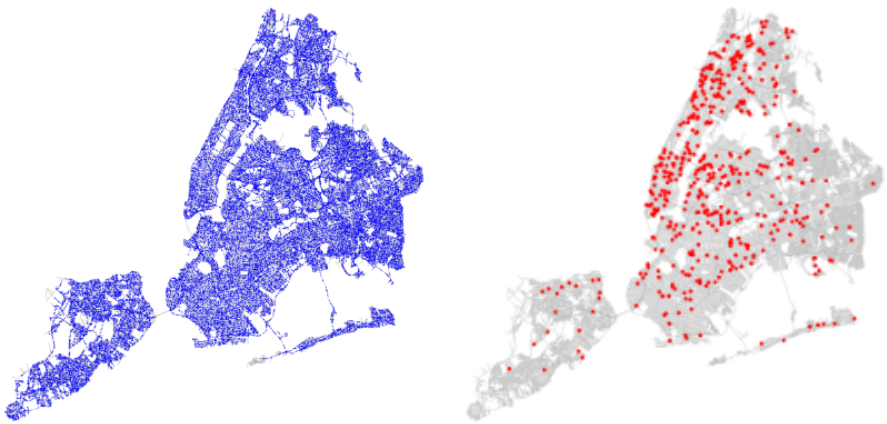
Figure 2: Examples of surveillance cameras in San Francisco. Upper left: Dome camera mounted on a traffic pole. Upper right: Dome camera mounted on the wall of a parking structure. Lower left: A wall mounted directed camera. Lower right: Two wall-mounted dome cameras.

Estimation

1. Avoid overcounting: Confirm all positives by humans
2. Count found cameras
3. Estimate number of cameras in entire city

$$\hat{K}_i = \frac{\#\text{camera_detections}}{\text{coverage} * \text{model_recall}}$$





e.g. New York: sample: 100′000, found: 556
⇒ Estimated density: 0.62 cameras/km

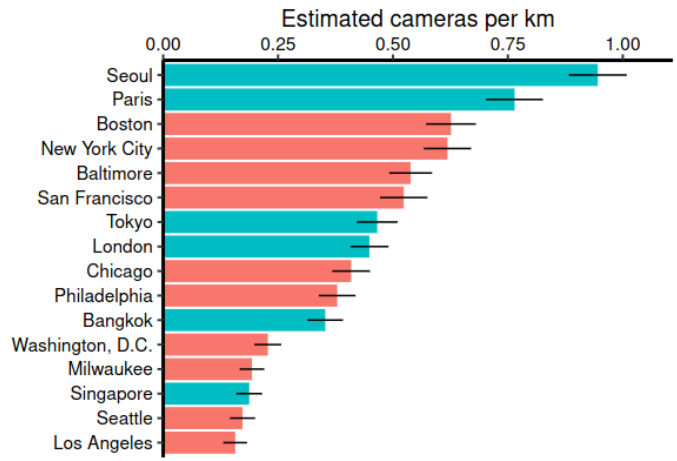
Results and Analysis

Analysis:

- Wide variety in density
- Higher density in mixed, commercial and industrial zones
- Lower in public and residential zones
- Higher density where high diversity

Restriction:

- Underestimation of total #cameras (no doorbell & indoor)
- No information about camera usage



My thoughts

- Would trust relative differences, not exact density numbers
- Estimation not dependent on government information
- Further research:
  - Other factors that might correlate with density (e.g. poverty)
  - Differences in density between US cities
  - In future: camera density development compared to city development

Thank you!

Discussion

Are there risks to private companies like Google possessing so much sensitive data?
In general: Is there a right balance between privacy and surveillance?